

CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Exp-03: Bubble Sort with Big-O Analysis

Aim:

To sort an array using Bubble Sort and analyze its time and space complexity using Big-O notation.

Algorithm:

1. Read the size of the array.
2. Read the array elements.
3. Compare adjacent elements.
4. Swap them if they are in the wrong order.
5. Repeat the process until the array is sorted.
6. Display the sorted array and complexity.

Source Code:

```
n = int(input("Enter number of elements: "))
arr = list(map(int, input("Enter array elements: ").split()))
for i in range(n - 1):
    for j in range(n - i - 1):
        if arr[j] > arr[j + 1]:
            arr[j], arr[j + 1] = arr[j + 1], arr[j]
print("Sorted Array:", arr)
print("Time Complexity: O(n^2)")
print("Space Complexity: O(1)")
```

Output:

```
Enter number of elements: 6
Enter array elements: 8 34 22 89 60 11
Sorted Array: [8, 11, 22, 34, 60, 89]
Time Complexity:  $O(n^2)$ 
Space Complexity:  $O(1)$ 
```

Result:

The given array was successfully sorted using Bubble Sort and its Big-O complexity was analyzed.